Introduction to spin geometry 2

Recap

Clifford algebra: $Cl(V,\langle,\rangle) := \bigoplus_{r\geq 0} V^{\otimes r} / \langle v\otimes v + |v|^2,\ v\in V\rangle$

V: Eucl. vector space

- $v\in V \Rightarrow v\cdot v = -|v|^2$
- $v \perp w \Rightarrow v\cdot w = -w\cdot v$
- $\dim V = n \Rightarrow \dim Cl(V) = 2^n$

$Cl_n := Cl(\mathbb{R}^n, \langle,\rangle_{Eucl.})$, $\mathbb{C}l_n := Cl_n \otimes_{\mathbb{R}} \mathbb{C}$

Classification of cplx. Clifford algebras

$\mathbb{C}l_{2k} \cong Mat(2^k,\mathbb{C})$, $\mathbb{C}l_{2k+1} \cong Mat(2^k,\mathbb{C}) \oplus Mat(2^k,\mathbb{C})$

Dirac bundle:

Hermition vector bundle $S\to M$ with

- Clifford module $\rho: Cl(TM) \to End_{\mathbb{C}}(S)$
- metric connection ∇^S on S s.t.

$$\nabla^S_X(v\cdot u) = (\nabla^{TM}_X v)\cdot u + v\cdot \nabla^S_X u$$

$\forall X,v\in C^\infty(TM)$, $u\in C^\infty(S)$

Dirac operator: $D = \sum_{i=1}^n e_i\cdot \nabla^S_{e_i}$

Weitzenböck formula: $D^2 = \Delta + \mathcal{K}$

Spin groups

Def: $Cl_n^\times \subset Cl_n$ mult. group of invertible elements
(each $0 \neq v \in \mathbb{R}^n$ lies in $Cl_n^\times$)

$$Spin(n) := \{ v_1 \cdot \ldots \cdot v_{2k} \in Cl_n^\times \mid v_1, \ldots, v_{2k} \in \mathbb{R}^n, |v_i| = 1, k \geq 0 \}$$

$v \in \mathbb{R}^n$, $|v| = 1$

- $x = \lambda v$, $\lambda \in \mathbb{R}$ $\Rightarrow$ $-v \cdot x \cdot \underbrace{v^{-1}}_{= -v} = -x$
- $x \perp v$ $\Rightarrow$ $-v \cdot x \cdot v^{-1} = x$

$\Rightarrow \Phi_v : \mathbb{R}^n \to \mathbb{R}^n$, $x \mapsto vxv^{-1}$ is negative of reflection at $v^\perp \subset \mathbb{R}^n$

$\rightsquigarrow$ extends to group homomorphism

$$\Phi : Spin(n) \to SO(n), \quad v \mapsto (x \mapsto \Phi_v(x) = v \cdot x \cdot v^{-1})$$

called adjoint representation of $Spin(n)$ on $\mathbb{R}^n$

Facts:

- $\ker(\Phi) = \{\pm 1\}$

  $\rightsquigarrow Spin(n) = \{ v_1 \cdot \ldots \cdot v_{2k} \mid 0 \leq 2k \leq n, |v_i| = 1 \}$
- $Spin(n)$ is a compact Lie-group
- Φ is two-fold covering map.
  In particular, $\dim Spin(n) = \frac{n(n-1)}{2}$
- $Spin(n)$ is connected for $n \geq 2$ and simply connected for $n \geq 3$

$$1 \to \mathbb{Z}/2 \longrightarrow Spin(n) \longrightarrow SO(n) \to 1$$

Example: • $Spin(1) = \mathbb{Z}/2$, $Spin(1) \xrightarrow{\Phi} SO(1) = \{1\}$

• $Spin(2) = \{ v_1 \cdot v_2 \mid v_1, v_2 \in S^1 \subset \mathbb{R}^2 \} \subset Cl(2) \cong \mathbb{H}$

$e_1 \mapsto i$, $e_2 \mapsto j$
$(e_1 e_2 \mapsto k)$

Let $v_1 = \alpha i + \beta j \in S^1 \subset span(i,j)$ $\quad |\alpha|^2 + |\beta|^2 = 1$

$v_2 = a i + b j \in S^1$

$\to v_1 \cdot v_2 = \underbrace{(-\alpha a - \beta b)}_{=\gamma} + \underbrace{(\alpha b - \beta a)}_{=z} k$

$\gamma^2 + z^2 = 1$

$\rightsquigarrow Spin(2) = S^1 \subset span(1,k)$

Let $\gamma + z k \in S^1$, $x i + y j \in \mathbb{R}^2$

$\rightsquigarrow \Phi_{\gamma + zk}(xi + yj) = (\gamma + zk)^2 (xi + yj)$

$\Rightarrow \Phi : S^1 \to S^1$, $z \mapsto z^2$

From Cliffor modules to Clifford bundle

W Cl_n-module s.t. Clifford mult. with unit vectors in $\mathbb{R}^n$ is an isometry, i.e. $|e| = 1$

$$\langle e \cdot u, e \cdot v \rangle_W = \langle u, v \rangle_W$$

$u, v \in W$

Goal: construct $CL(TM)$-module bundle $S \to M$ with typical fibre isometric to W

$$\left(S = \begin{array}{c} CL(TM) \\ \| \\ \Lambda^* TM \end{array} \right)$$

(M, g) oriented Riem. mfd.

Choose trivialization covering $\Psi_i : TM|_{U_i} \xrightarrow{\sim} U_i \times \mathbb{R}^n$

s.t.

- $\Psi_{ji} = \Psi_j \circ \Psi_i^{-1} : U_i \cap U_j \longrightarrow SO(n)$
- the intersection of finitely many U_i is empty or contractible

Need to find transition maps

$$\rho_{ji} : U_i \cap U_j \longrightarrow Aut(W)$$

s.t.

- cocycle condition

  $\forall ijk$: $\rho_{ki} = \rho_{kj} \circ \rho_{ji}$ on $U_i \cap U_j \cap U_k$

- Compatible with Clifford mult.

  $\forall ij$, $\forall x \in U_i \cap U_j \cap U_k$, $\forall v \in \mathbb{R}^n$ and $\forall w \in W$

$$(\Psi_{ji})_x(v) \cdot (\rho_{ji})_x(w) = (\rho_{ji})_x(v \cdot w)$$

For $x \in U_i \cap U_j$ let $\xi \in Spin(n)$ s.t. $\Phi(\xi) = (\Psi_{ji})_x \in SO(n)$

set $(\rho_{ji})_x(w) := \xi \cdot w$

$$\Rightarrow \quad (\psi_{ji})_x(v)\cdot(\rho_{ji})_x(w) = \left(\lambda\cdot v\cdot\lambda^{-1}\right)\cdot(\lambda\cdot w) = \lambda\cdot v\cdot w$$

$$= (\rho_{ji})_x(v\cdot w)$$

$$\begin{array}{ccc} & & Spin(n)\subset Cl_n \longrightarrow Aut(W) \\ & \exists\rho_{ji} \nearrow & \downarrow \Phi \\ U_i\cap U_j & \xrightarrow[\psi_{ji}]{} & SO(n) \end{array}$$

Set $\quad b_{ijk} = \rho_{ki}^{-1}\rho_{kj}\rho_{ji} : U_i\cap U_j\cap U_k \longrightarrow Spin(n)$

$\Phi(b_{ijk}) = 1$ as ψ_{ji} satisfy cocycle cond.

$\rightsquigarrow$ $im(b_{ijk}) \subset \mathbb{Z}/2 = \ker(\Phi)$

$\rightsquigarrow$ (b_{ijk}) defines cochain $b\in\check{C}^2(M,\mathbb{Z}/2)$

$\rightsquigarrow$ $db = 0$

$\rightsquigarrow$ $w_2(M) = [b]\in H^2(M,\mathbb{Z}/2)$ <u>2nd Stiefel-Whitney class</u>

If $w_2(M)=0$, then $\exists\tau\in C^1(M,\mathbb{Z}/2)$ s.t. $d\tau = b$

$\tilde{\rho}_{ji} := \rho_{ji}\tau_{ji}$

$\rightsquigarrow$ $\tilde{\rho}_{ji}$ satisfy cocycle cond.

<u>Def</u> • M <u>spinable</u> if $w_2(M)=0$ (M orient $\Leftrightarrow w_1(M)=0$)

• A <u>Spin-structure</u> on Riem. spin. mfd. (M,g) is a collection (U_i,ψ_i,ρ_{ij})

U_i: open cover of M; ψ_i: orthog. orient.-preserv. local trivial. of TM; ρ_{ij}: lifts of ψ_{ji}

- (M,g) Riem spin mfd., W Cl_n-module

  $\Rightarrow$ $S \to M$ constructed above is called spinor bundle associated with W

Connections on spinor bundles

Let

- $TM|_U \overset{(*)}{\cong} U \times \mathbb{R}^n$, $S|_U \cong U \times W$ as above
- $(e_1, \dots, e_n)$ standard ON-frame of $U \times \mathbb{R}^n$
- $so(n) = \{A \in \mathbb{R}^{n \times n} \mid A = -A^T\}$ Lie alg. of $SO(n)$
- $\omega = (\omega_i^j) \in \Omega^1(U, so(n))$ connection form of ∇^{TM} w.r.t. $(*)$

  i.e.

$$\nabla_X^{TM} e_i = \omega(X)(e_i) = \sum_{j=1}^{n} \omega_i^j(X) e_j \in C^\infty(U, \mathbb{R}^n)$$

Def: Let $w \in C^\infty(U, W)$ be a const. section

Define:

$$\nabla_X^S w = \frac{1}{2} \sum_{1 \le i < j \le n} \omega_i^j(X) \cdot e_i \cdot e_j \cdot w \in C^\infty(U, W)$$

Fact: this defines a global connection on S which turns $S \to M$ into a Dirac bundle

∇^S is called spinor connection

Thm (Schrödinger–Lichnerowicz)

$S \to M$ spinor Dirac bundle for some Cl_n-module W

The corresp. Dirac operator satisfies:

$$D^2 = \Delta + \frac{1}{4} scal_g$$

Twisted spinor bundles:

Let $E \to M$ be a Hermitian vector bundle with compatible connection ∇^E

Set $\nabla^{S\otimes E} := \nabla^S \otimes id + id \otimes \nabla^E$

$\rightsquigarrow (S\otimes E, \nabla^{S\otimes E})$ is again a Dirac bundle s.t.

$$D_E^2 = \Delta + \frac{1}{4} scal_g + \underbrace{\mathcal{R}^E}_{= \sum_{i<j} e_i \cdot e_j R^E(e_i, e_j)}$$

$\uparrow$ curv. op. of (E, ∇^E)